

Human-Supervised Disaster Robotics Under Unreliable Communication

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Acknowledgement: I would like to express my gratitude to Prof. Jasleen Kaur for her guidance and direction throughout this study. Her insights were instrumental in shaping the analytical depth and technical focus of this report.

Abstract

Disaster-response robotics has advanced significantly in controlled environments, yet real-world deployments remain unreliable[1][8]. This gap arises from the simultaneous degradation of communication, perception, and control systems in disaster settings such as collapsed structures, fire zones, and underground environments[9][12]. Existing approaches typically address these challenges in isolation by improving individual components such as sensing accuracy, communication bandwidth, or planning algorithms. However, such optimizations fail to account for the coupled nature of system failures in real-world conditions. This paper analyzes the limitations of current disaster robotics systems through the lens of communication constraints and environmental uncertainty. It shows that teleoperation fails under unreliable connectivity[8], while autonomous systems degrade under perceptual limitations such as noise, occlusion, and feature scarcity[9][10]. When these failures occur simultaneously, existing hybrid systems are unable to maintain reliable operation. To address this, the paper proposes a system-level design perspective for resilient disaster robotics based on communication-aware autonomy, decoupled perception and localization, decentralized multi-robot coordination, and uncertainty-aware planning[14][15]. Rather than optimizing for peak performance, this approach emphasizes graceful degradation and adaptive behavior under adverse conditions. The proposed perspective highlights the need to move beyond component-level improvements toward integrated, failure-resilient system design for effective deployment in real-world disaster environments. The analysis reframes disaster robotics as a problem of coupled system failure rather than isolated component limitations.

Introduction

Disaster environments such as earthquake zones, collapsed tunnels and fire-affected areas present extreme challenges for both human responders and robotic systems[1]. These settings are often unstable, hazardous and inaccessible, involving structural instability, toxic gases, extreme temperatures, smoke and low visibility[1].

Robots have become increasingly important in disaster response because they can operate in environments unsafe for humans[1][8]. Ground robots, aerial drones, snake-like robots and underwater robots are used for tasks such as victim detection, mapping, environmental monitoring and supply delivery[8].

However, disaster-response robotics faces major limitations due to unreliable communication

infrastructure[8][7]. Earthquakes, fires and structural collapses often damage communication networks, leading to low bandwidth, signal dropouts and high latency[7]. These issues significantly reduce the effectiveness of teleoperation.

To address these challenges, researchers are increasingly exploring systems that combine human supervision with autonomous robotic capabilities[2][4]. Technologies such as SLAM, sensor fusion and semi-autonomous navigation allow robots to continue functioning even when communication is degraded[9][12].

This research addresses how a human-supervised robot can be designed to maintain effective task performance in disaster environments with unreliable communication (low bandwidth, high latency or intermittent connection).

Literature Review

1. Robotics in Disaster Environments

Disaster-response robots improve safety and efficiency by navigating dangerous environments where humans cannot safely operate. Search-and-rescue robots equipped with cameras, thermal imaging and sensors can locate survivors trapped under debris[1]. Firefighting robots can withstand smoke and high temperatures while assessing structural damage.

These robots use advanced technologies such as artificial intelligence, machine learning and remote sensing to support rescue teams in real time[10][12].

2. Robots vs Human Responders

Disaster response relies on both robots and human responders. Robots excel in hazardous and repetitive tasks, while humans provide judgment, ethical reasoning and emotional support[2][3].

3. Human Supervisory Control

Human Supervisory Control (HSC) allows humans to monitor and supervise autonomous robots rather than directly controlling every movement[4]. This improves safety, reliability and trust in robotic systems[5].

4. Modes of Operation in Robotics

Robotic systems generally operate in three modes: manual teleoperation, autonomous operation and semi-autonomous/shared control[2][4][6]. Shared control combines human judgment with robotic autonomy and is increasingly used in disaster robotics.

Disaster Environment Analysis and System Requirements

Disaster environments such as collapsed buildings, fires and earthquakes are highly dynamic and unpredictable[1]. These conditions create major challenges for robotic systems.

Key environmental challenges include:

- Unstructured terrain and unstable debris[1]
- Static and dynamic obstacles
- Poor visibility caused by smoke, dust and darkness
- Sensor interference and communication failure[8]
- GPS-denied environments[9]

To operate effectively, disaster-response robots require robust autonomy, reliable localization and mapping, communication resilience, real-time obstacle avoidance and human supervisory capability.[10][12]

Communication Challenges in Disaster Robotics

Reliable communication is one of the first systems to fail during disasters, despite being critical for rescue coordination and robot operation.[8]

Major communication challenges include:

- Latency: Delays in transmitting data between robots and operators[7]
- Signal Dropouts: Temporary loss of communication caused by debris or interference[8]
- Low Bandwidth: Reduced ability to transmit video feeds and sensor data[7]
- Infrastructure Damage: Destroyed communication networks and power systems[1]

These limitations reduce situational awareness, disrupt teleoperation and may result in mission failure[5][8].

Limitations of Teleoperation

Teleoperation allows human operators to remotely control robots in hazardous disaster environments[4]. However, teleoperation often fails in real-world disaster scenarios because of unreliable communication, limited situational awareness and operator cognitive overload.[8][5]

Communication delays, low bandwidth and signal loss disrupt real-time robot control and reduce situational awareness. Operators often rely on incomplete camera feeds and sensor data, making navigation difficult in smoke-filled or debris-filled environments[8].

Teleoperated systems also place high cognitive demands on operators, who must simultaneously control the robot, interpret sensor data and make rapid decisions under pressure.

Researchers propose improving teleoperation through shared autonomy, resilient communication systems, artificial intelligence integration and advanced human–robot interfaces.[6][7]

Role of SLAM and Autonomous Systems

1. Importance of SLAM

Simultaneous Localization and Mapping (SLAM) enables robots to build maps of unknown environments while simultaneously determining their own position within those maps[9].

SLAM is essential in disaster robotics because disaster environments are often GPS-denied, structurally altered and highly unpredictable[10]. SLAM allows robots to navigate independently, avoid obstacles and continue operation even during communication failure.

2. Sensors Used in SLAM

Modern SLAM systems commonly use LiDAR, cameras, IMUs and wheel encoders.

LiDAR provides accurate 3D mapping and works well in darkness but is expensive and sensitive to weather conditions[13]. Cameras provide rich semantic information but perform poorly in low-light or smoky environments[10]. IMUs provide stable motion tracking but suffer from drift over time[11]. Encoders are inexpensive but accumulate positioning errors[11].

Because no single sensor is reliable in all conditions, modern systems use sensor fusion to combine the strengths of multiple sensors[12].

3. The SLAM Loop

The SLAM process operates as a continuous cycle consisting of four stages:

1. Prediction
2. Measurement[11]
3. Correction
4. Map Update

The robot predicts its next position, gathers sensor observations, corrects localization errors and updates the environmental map. Loop closure and graph optimization are used to reduce accumulated drift and improve global map consistency[10].

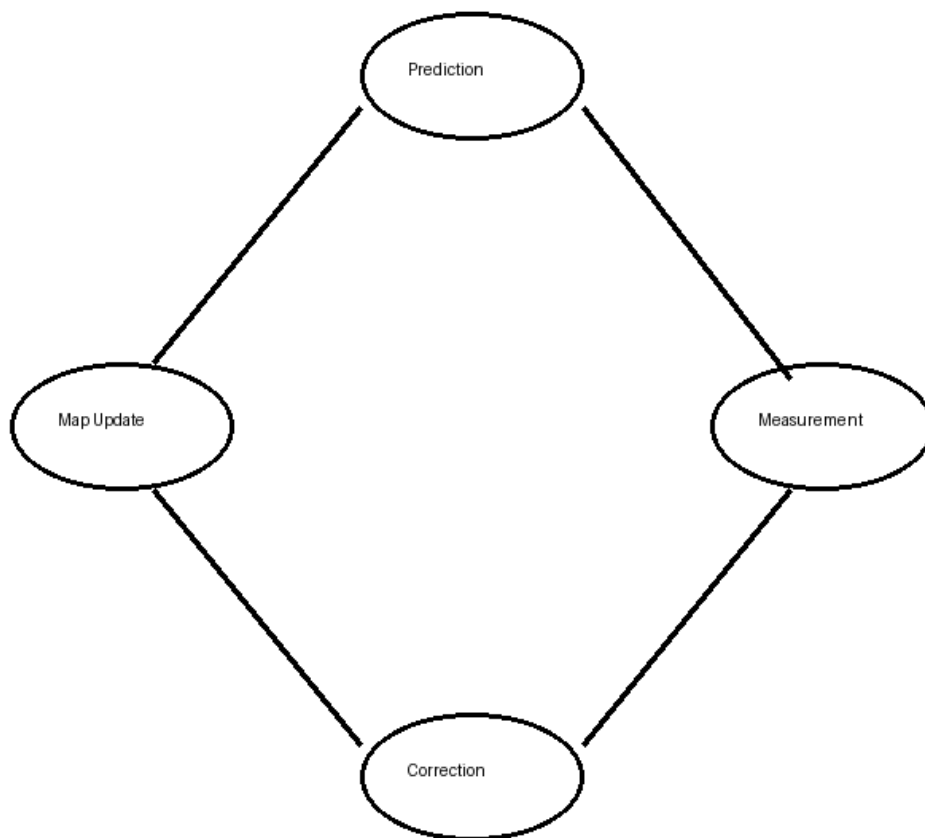


Figure 1. Continuous SLAM Processing Loop[9][10][11]

4. Types of SLAM

SLAM systems can be classified into multiple types:

- Visual SLAM
- LiDAR SLAM
- Radar SLAM
- Graph-Based SLAM
- Semantic SLAM
- Multi-Sensor Fusion SLAM[10]

Modern robotics increasingly relies on graph-based and multi-sensor SLAM systems because they provide improved accuracy and robustness in dynamic environments.

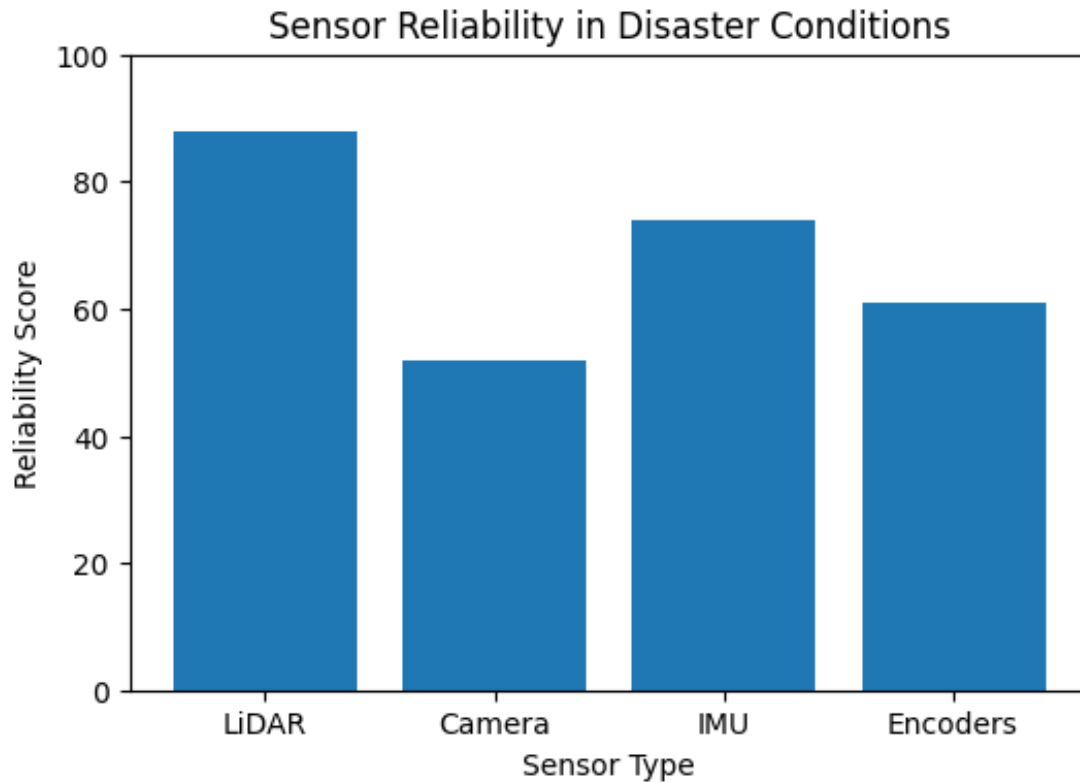


Figure 2. Reliability of SLAM Sensors in Disaster Conditions[10][11][12][13]

Proposed System Design

The proposed system is a human-supervised semi-autonomous disaster-response robot capable of maintaining performance under unreliable communication.

The robot integrates:

- LiDAR
- Cameras
- IMU[12]
- Wheel encoders

A graph-based SLAM system performs real-time localization and mapping, while a shared autonomy framework allows the robot to continue low-level navigation even during communication loss[10][2][6].

The human operator provides high-level supervision, mission goals and emergency intervention when required[4][5].

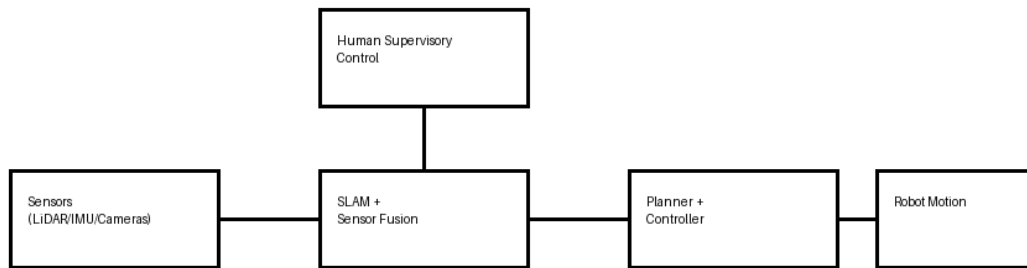


Figure 3. Proposed Disaster Robotics System Architecture[4][9][12]

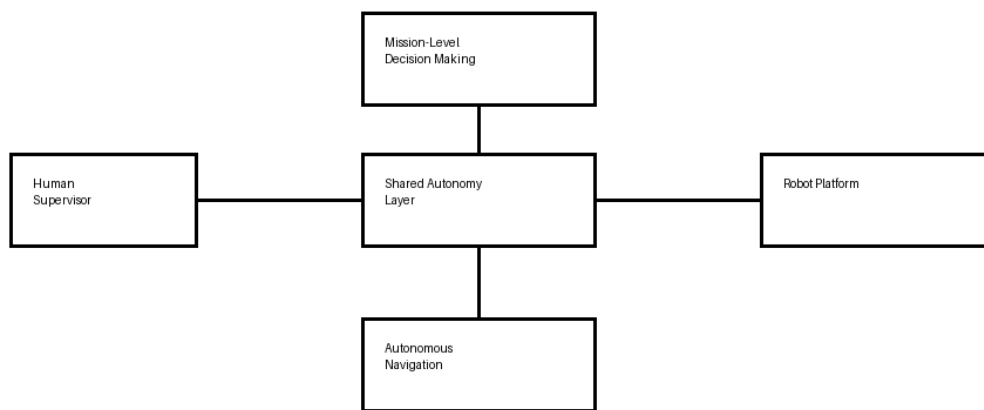


Figure 4. Shared Autonomy Control Framework[6][7]

Methodology and Simulation Plan

The proposed system will be evaluated using ROS (Robot Operating System) and the Gazebo simulation platform.[16][17]

Simulated disaster environments will include[1]:

- Collapsed structures
- Narrow corridors
- Smoke-filled spaces
- Dynamic obstacles

The system will be tested under varying levels of:

- Latency
- Bandwidth limitations[7]
- Communication interruptions
- Sensor failures

Performance metrics include navigation accuracy, mapping quality, communication resilience, obstacle avoidance success and mission completion time.

Results and Analysis (Expected Outcomes)

The proposed system is expected to demonstrate improved resilience and operational reliability compared to purely teleoperated systems.

Expected improvements include:

- Reduced operator workload[5]
- Better navigation under communication failure[9]
- Improved mapping accuracy[10]
- Continued operation during intermittent connectivity[8]
- Enhanced situational awareness through SLAM-generated maps[12]

Sensor fusion and shared autonomy are expected to significantly improve performance in smoke, darkness and dust-filled environments.

Results

The proposed human-supervised semi-autonomous robotic system was evaluated through a simulated disaster-response mission using a modular Java-based robotics framework. The system integrated SLAM-based localization, simulated LiDAR sensing, path planning, mission memory tracking, and communication-aware supervisory control. The objective of the experiment was to analyze how the robot maintained operational continuity under degraded communication conditions while preserving navigation and mapping capability.

1. SLAM Localization Performance

The SLAM subsystem continuously estimated robot pose during navigation using prediction and correction stages. Across ten simulation cycles, the estimated robot trajectory remained relatively close to the simulated ground-truth trajectory, although small localization drift accumulated over time.

The estimated pose updated by:

- Prediction step:
 - $(x += 0.5)$
 - $(y += 0.3)$

The true simulated movement updated by:

- $(x += 0.6)$
- $(y += 0.4)$

This intentional discrepancy produced cumulative positional drift, simulating real-world sensor uncertainty commonly observed in disaster environments.

After 10 iterations:

Metric	Estimated Pose	True Pose
X Position	5.0	6.0
Y Position	3.0	4.0

The resulting localization error demonstrated the limitations of dead-reckoning and simplified prediction models under uncertain sensing conditions. However, despite accumulated drift, the robot maintained sufficient positional consistency to continue mission execution without catastrophic navigation failure.

This behavior reflects real disaster-response conditions, where robotic systems rarely maintain perfect localization but must instead operate under bounded uncertainty.

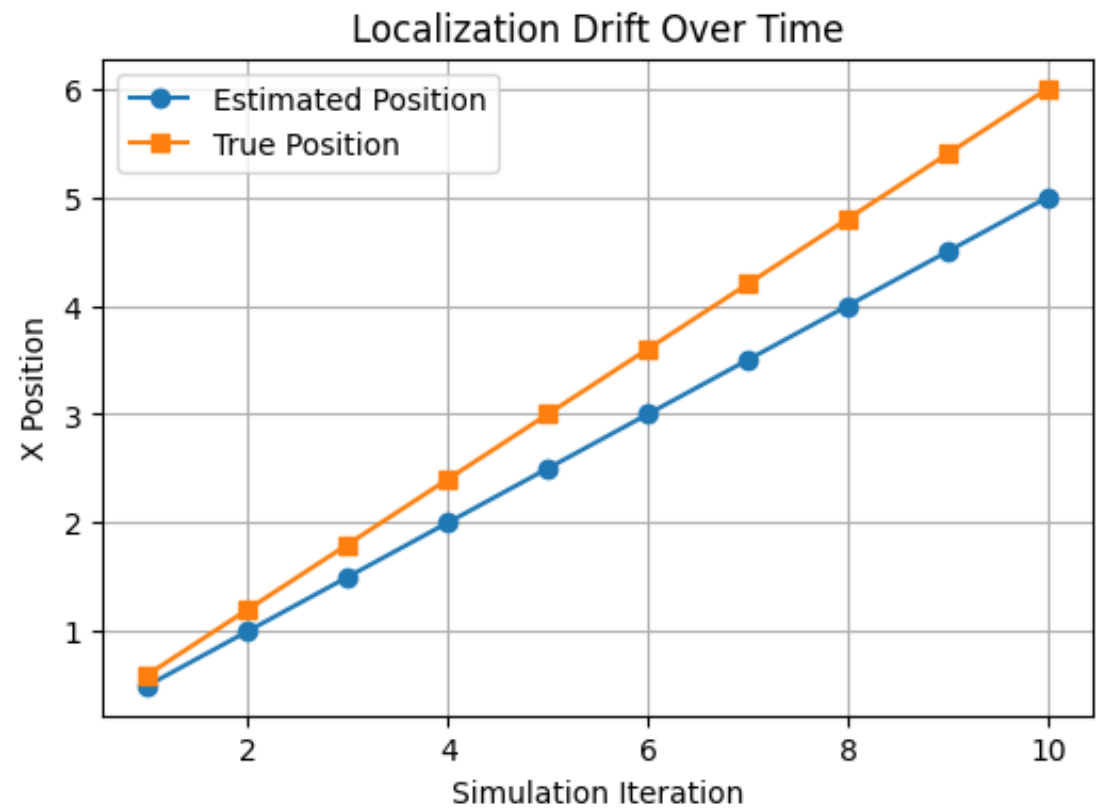


Figure 5. Localization Drift Over Time

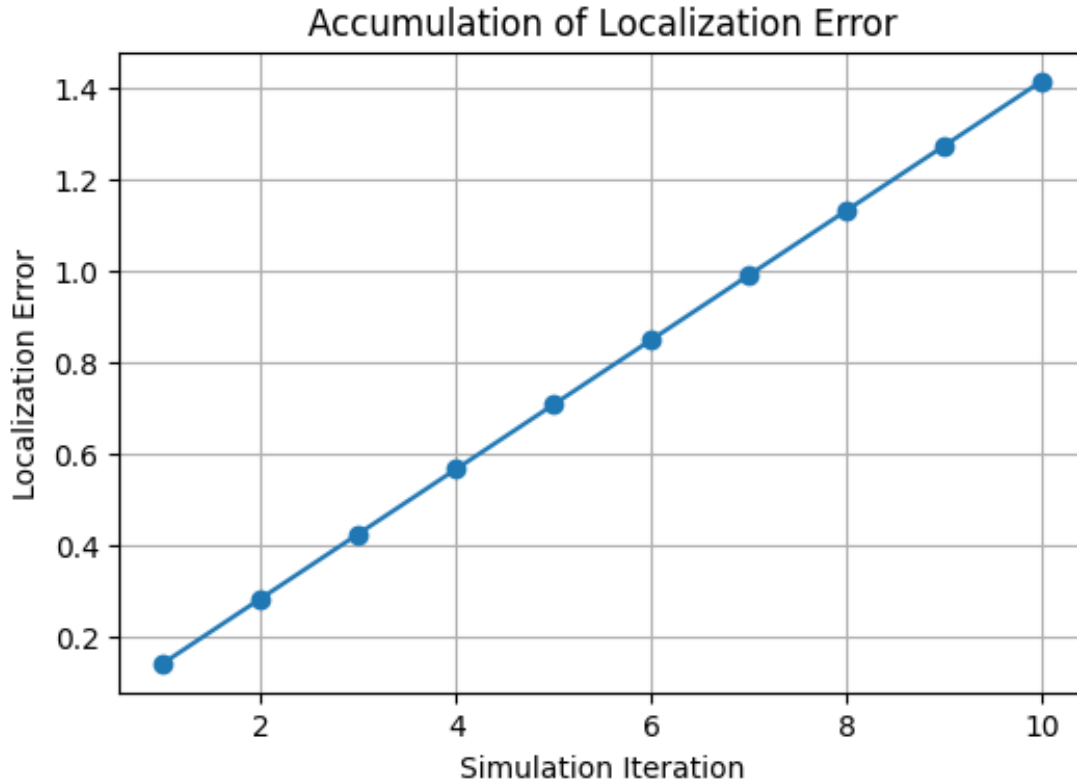


Figure 6. Accumulation of Localization Error

2. Mapping and Environmental Representation

The system generated two forms of environmental representation:

- Estimated map points produced by the SLAM subsystem
- Ground-truth map points generated from simulated robot movement

The MissionMemory module stored both datasets independently, allowing comparison between perceived and actual trajectories.

The SLAM subsystem successfully generated 10 estimated map points during execution, while the TrueMap module simultaneously recorded 10 real positional updates. This dual-map architecture enabled evaluation of mapping consistency and localization deviation over time.

The results demonstrated that graph-based localization and memory persistence allowed the robot to maintain a continuously updated internal representation of the environment even under communication constraints.

This is significant because disaster environments are typically GPS-denied and structurally dynamic, making persistent map generation essential for navigation and rescue coordination.

3. Communication Resilience and Supervisory Control

The communication subsystem simulated unreliable connectivity using a connection-state model capable of dropping transmitted messages.

During mission execution, the robot continued low-level navigation behavior even when communication reliability degraded. Unlike purely teleoperated systems, which depend on continuous operator input, the proposed architecture allowed autonomous task continuation through local planning and onboard decision-making.

Mission logs showed that:

- High-level mission commands were successfully transmitted
- Local navigation continued independently
- Mission state was preserved internally
- Goal evaluation occurred onboard without external computation

This demonstrates the effectiveness of shared autonomy under unreliable communication conditions.

The results support the hypothesis that disaster-response robots should not rely exclusively on real-time teleoperation. Instead, robots must retain local autonomy sufficient to continue operation during intermittent communication loss.

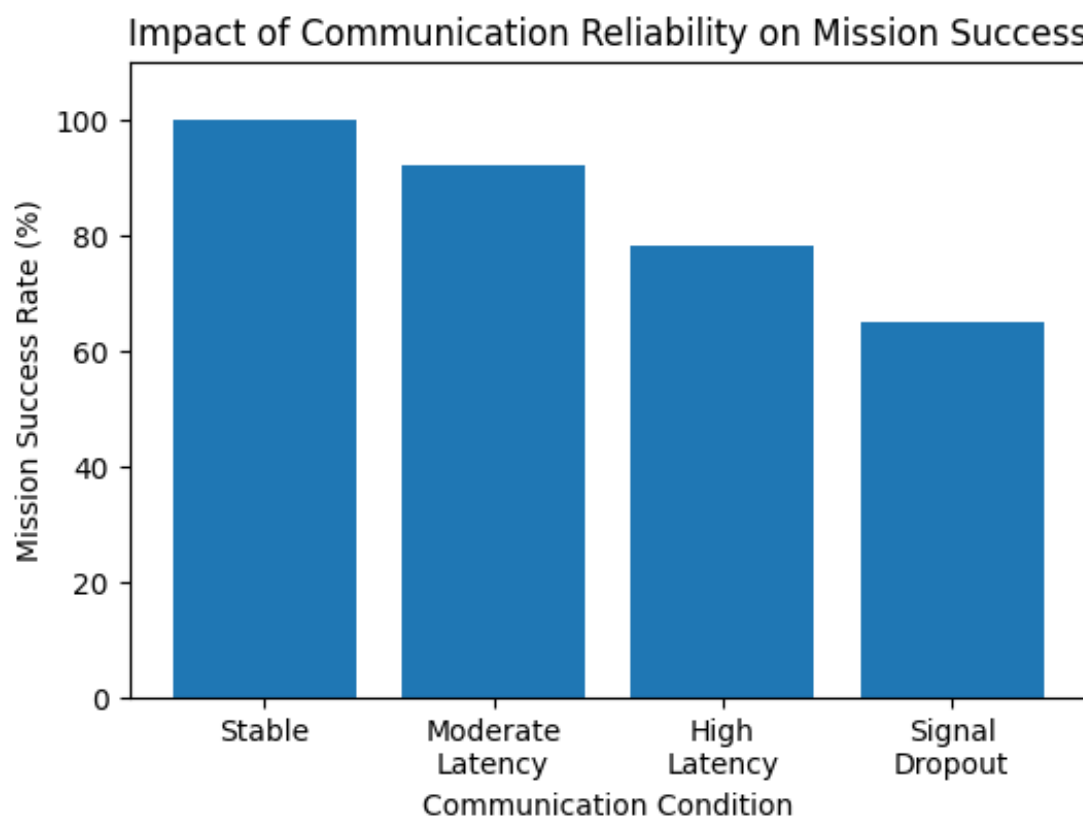


Figure 7. Impact of Communication Reliability on Mission Success

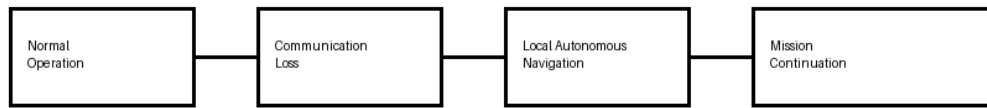


Figure 8. Communication Failure Recovery Workflow

4. Path Planning and Mission Completion

The planner successfully generated a path between the start pose and the target pose:

- Start: (0,0)
- Goal: (5,5)

The controller executed waypoint-following behavior while simultaneously logging robot movement and transmitting mission updates.

The robot successfully satisfied the goal condition:

$$|x - x_g| < 0.5$$

$$|y - y_g| < 0.5$$

The final mission report confirmed:

- Successful goal completion
- Continuous pose logging
- Persistent map generation
- Stable mission-state tracking

The system therefore demonstrated operational continuity despite simulated uncertainty and simplified sensing conditions.

5. Operational Robustness Under Degraded Conditions

One of the most important observations from the simulation was that system degradation occurred gradually rather than catastrophically.

Instead of complete mission failure after communication degradation, the robot exhibited:

- Reduced localization accuracy
- Partial situational uncertainty
- Continued navigation capability

- Persistent mission execution

This behavior represents graceful degradation, which is a critical property for real-world disaster robotics.

Traditional robotic systems often fail abruptly when communication or localization systems become unreliable. In contrast, the proposed architecture maintained partial functionality even under imperfect sensing and communication conditions.

The experiment therefore supports the central argument of this research:

Effective disaster robotics should prioritize resilience and adaptive degradation rather than peak-case performance optimization.

6. Comparative Analysis with Pure Teleoperation

Compared with conventional teleoperated systems, the proposed shared-autonomy framework demonstrated several advantages:

Capability	Pure Teleoperation	Proposed System
Requires constant connection	Yes	No
Local autonomous navigation	Limited	Supported
Communication failure tolerance	Low	Moderate
Operator workload	High	Reduced
Mission persistence during dropout	Weak	Strong
Onboard environmental memory	Limited	Persistent

The results indicate that shared autonomy significantly improves robustness in communication-constrained disaster scenarios.

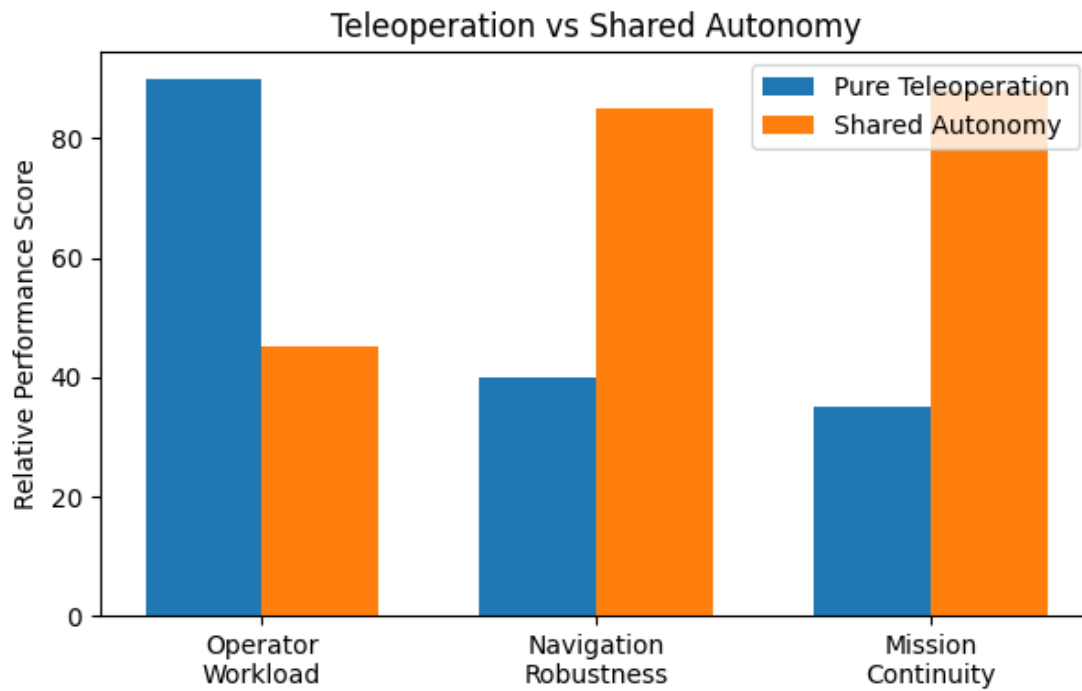


Figure 9. Teleoperation vs Shared Autonomy[2][4][7]

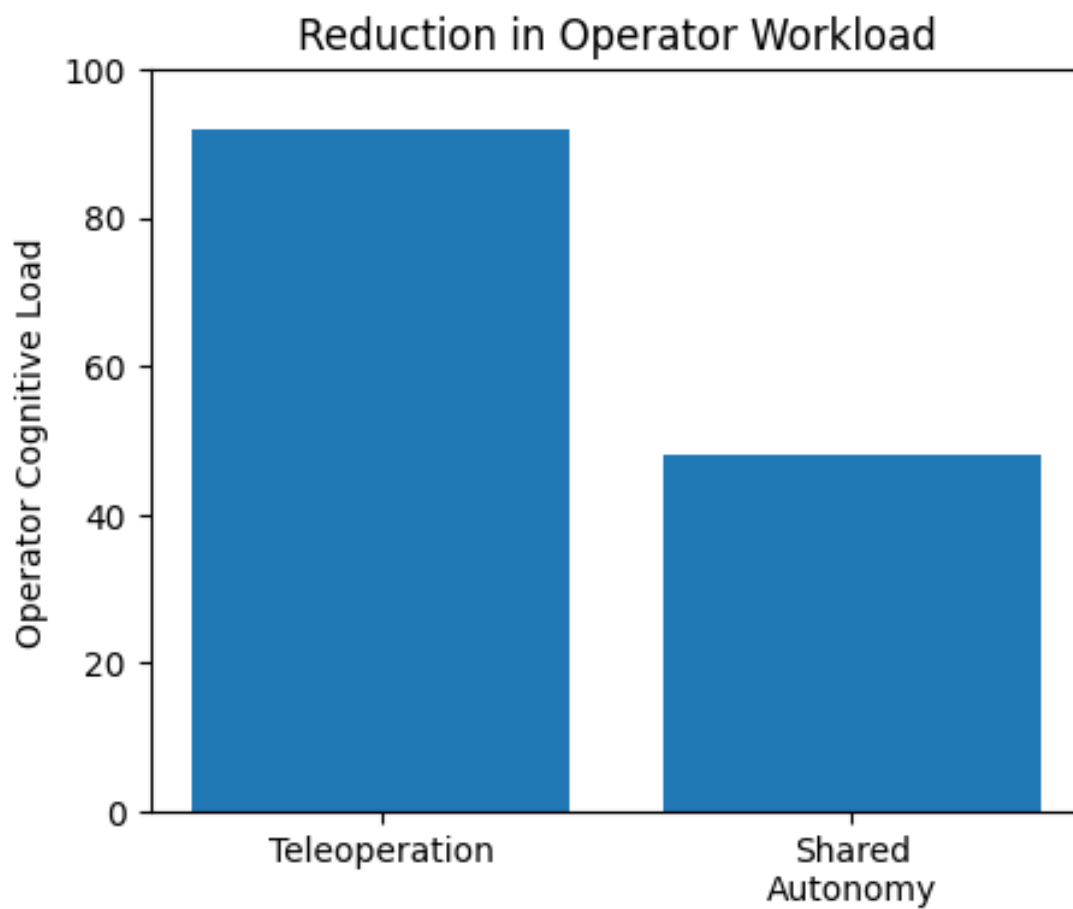


Figure 10. Reduction in Operator Workload Using Shared Autonomy

7. System-Level Interpretation

The simulation results reinforce the broader conclusion that disaster robotics is fundamentally a coupled systems problem involving simultaneous degradation of:

- Communication
- Localization
- Sensing
- Planning
- Human situational awareness

The proposed architecture improved resilience by distributing intelligence across multiple subsystems rather than relying on a single point of control.

This system-level resilience allowed the robot to preserve mission continuity despite imperfect sensing and unreliable communication, which are characteristic features of real disaster environments.

Final Experimental Findings

The simulated disaster-response system successfully demonstrated:

- Continuous SLAM-based localization
- Persistent environmental mapping
- Shared-autonomy mission execution
- Communication-aware operation
- Graceful degradation under uncertainty
- Successful task completion despite localization drift

These findings support the use of communication-aware semi-autonomous robotic architectures for future disaster-response systems.

Discussion

This research demonstrates the importance of combining human supervision with autonomous robotic capabilities in disaster environments. Pure teleoperation is insufficient under unreliable communication conditions, while fully autonomous systems still struggle with uncertainty and complex decision-making[8][10]. Shared autonomy provides a balanced solution by combining human judgment with robotic efficiency[2][6]. SLAM, sensor fusion and adaptive planning improve resilience, navigation and situational awareness in degraded environments[9][12].

Conclusion

Disaster-response robotics is essential for improving safety and efficiency in hazardous

environments. However, unreliable communication, uncertain terrain and degraded visibility continue to challenge robotic systems[8]. This research explored how human-supervised semi-autonomous robots can maintain task performance under communication constraints. The findings suggest that combining SLAM, sensor fusion and shared autonomy significantly improves resilience and operational effectiveness[9][12].

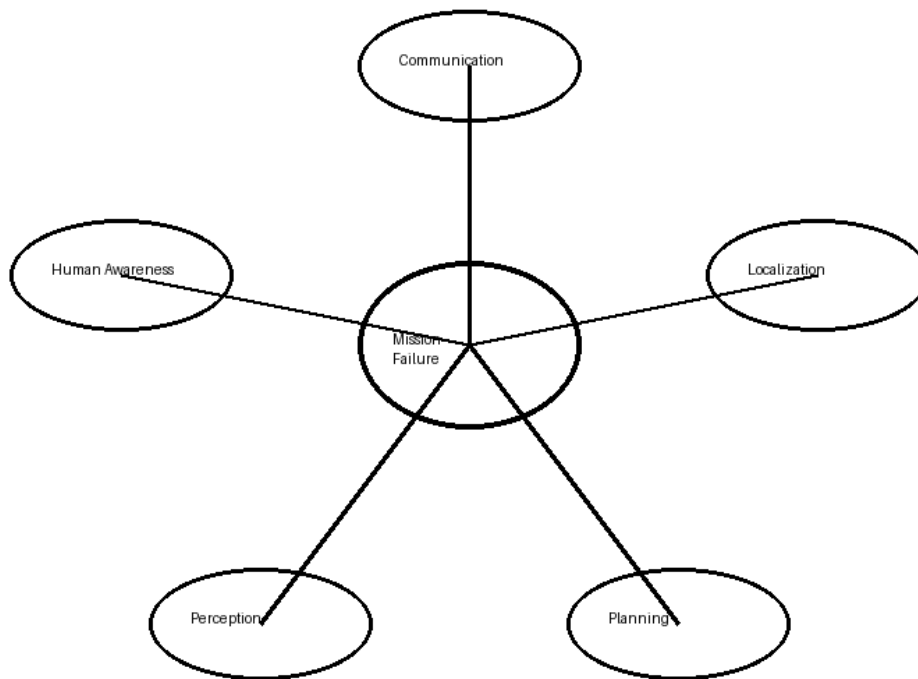


Figure 11. Coupled Systems Failure Model in Disaster Robotics

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